VIGNESH J

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Summary

– ECE graduate with hands-on full-stack experience (Python/FastAPI, Node.js/Express, REST APIs, SQL/NoSQL) and applied GenAI experience (LLM integration, prompt engineering, RAG with Gemini). Comfortable across the full SDLC and eager to grow into production-grade fullstack & GenAI engineering.

Education

Vellore Institute of Technology – Chennai **Sep 2022 – Jun 2026**
B.Tech in Electronics and Communication Engineering (CGPA: 8.61/10) *Tamil Nadu, India*

Technical Skills

– **Languages:** Python, Java, JavaScript, TypeScript, Kotlin, C, C++, Dart
– **Frontend:** HTML, CSS, JavaScript, TypeScript, ReactJS (certified)
– **Backend & APIs:** FastAPI, REST APIs, Node.js, Express.js, Flask
– **Databases:** SQL (queries, joins, aggregations, subqueries), PostgreSQL, MySQL, MongoDB, SQLite (Room ORM), Firebase Firestore (NoSQL)
– **GenAI / AI-ML:** LLM Integration (Gemini, Phi-4-mini), Prompt Engineering, RAG, YOLOv11, TinyML
– **Cloud & DevOps:** AWS (EC2, Cloud Practitioner), Firebase, Git, GitHub, GitHub Actions, Docker (basics)
– **Practices:** SDLC, Agile, Code Reviews, Unit Testing, Technical Documentation, Data Structures & Algorithms (100+ LeetCode)

Experience

Annam.ai – Center of Excellence in AI for Agriculture **May 2025 – Jul 2025**
Python Developer & Machine Learning Engineering Intern *IIT Ropar, Punjab*
– Fine-tuned and deployed a YOLOv11 real-time insect-detection model, serving it through REST APIs built with FastAPI (Python).
– Built Python data pipelines (Pandas, NumPy) for agricultural data ingestion and cleaning; prepared technical documentation for cross-functional stakeholders.
– Integrated Gemini 2.5 API and Perplexity Sonar LLM into an AI-powered crop advisory chatbot using prompt engineering and RAG pipelines.
– Developed a Flutter frontend with IoT sensor and OpenWeather API integration for live agri-data tracking and farmer alerts.

Projects

DeskMind – On-Device AI Lecture Assistant | *Python, Kotlin, llama.cpp, SQLite, Git* | GitHub **2025 – 2026**
– Designed and built a fully offline Android AI app end-to-end (requirements, design, implementation, testing, deployment) following SDLC practices.
– Integrated Phi-4-mini LLM via llama.cpp JNI for on-device note structuring with a 4-pass pipeline (segmentation, elaboration, correction, to-do extraction).
– Wrote 22 unit tests and implemented GitHub Actions CI/CD to automate testing on every push; used SQLite (Room ORM) for persistent storage.

ResQ5 – Traffic Signal Preemption *(Patent Pending)* | *Node.js, AWS EC2, Flutter, Firebase* | GitHub **Apr 2025**
– Field-deployed as a live pilot with the **Tambaram City Police Commissionerate**; filed a patent (final stage) for the underlying IoT-based signal-preemption system.
– Deployed and hosted a Node.js backend on AWS EC2, exposing REST APIs for real-time traffic-light control and ambulance tracking.
– Built a Flutter client with Google Maps API for live route visualization; used Firebase for real-time data sync and authentication.

Quiz Management System | *ReactJS, Node.js, Express.js, MongoDB* | GitHub | Live Demo **2024**
– Built a full-stack **MERN** quiz platform with JWT authentication and role-based access control for student and admin roles.
– Designed MongoDB schemas and RESTful APIs; implemented real-time quiz sessions with live scoring and an admin analytics dashboard; deployed on Render.

Certifications & Profile Links

HackerRank Python (Basic) – HackerRank | **React Developer Certification** – Ethnus (Jun 2024) | **Flutter & Dart – Complete Guide** – Udemy (May 2025)
LeetCode – 100+ problems solved in Data Structures & Algorithms